I. Fundamentals of Energy and Processes

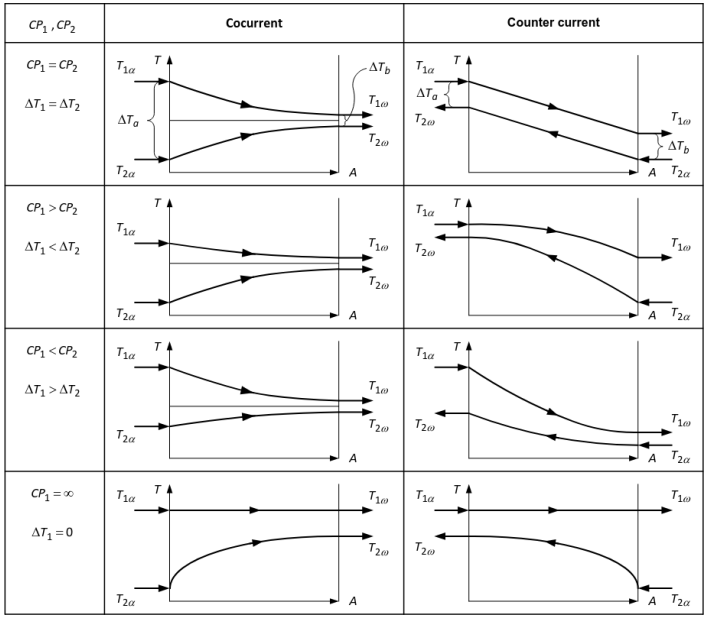
Heat exchanger

Energy balance

$$\dot{Q} = \dot{m} c_p \Delta\vartheta = CP\Delta\vartheta, \quad \dot{Q} = ¢$$

$$\dot{m}_1 c_{p1} \Delta\vartheta_1 = \dot{m}_2 c_{p2} \Delta\vartheta_2 \Longleftrightarrow CP_1 \Delta\vartheta_1 = CP_2 \Delta\vartheta_2$$

Average temperature difference



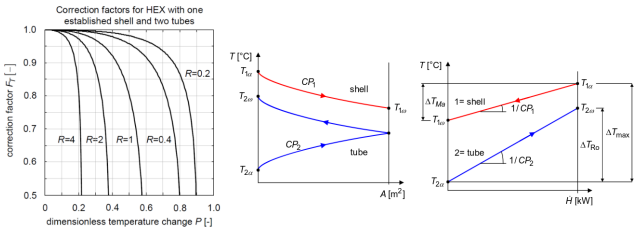
Logarithmic mean temperature difference (LMTD)

$$\Delta T_m = \frac{\Delta T_a - \Delta T_b}{\ln \frac{\Delta T_a}{\Delta T_b}}$$

$$\Delta T_a = |\vartheta_{1,a} - \vartheta_{2,a}| \quad ; \quad \Delta T_b = |\vartheta_{1,b} - \vartheta_{2,b}|$$

Corrected mean temperature heat flow

$$\dot{Q} = kA\Delta T_m \cdot F_T \text{, with } 0 < F_T < 1$$



Co-current vs countercurrent flow heat exchanger

$$\dot{Q} = kA_{\text{co}}\Delta T_{m,\text{co}} = kA_{\text{count}}\Delta T_{m,\text{count}}$$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\Delta T_{m,\text{count}}}{\Delta T_{m,\text{co}}}$$

$$\frac{A_{\text{co}}}{A_{\text{count}}} = \frac{\dot{Q}_{\text{co}}}{\dot{Q}_{\text{count}}} \cdot \frac{\Delta T_{m,\text{count}}}{\Delta T_{m,\text{co}}}, \quad \text{if } \dot{Q}_1 \neq \dot{Q}_2$$

Heat capacity rate ratio (Capacity ratio)

It measures the relative ability of stream 1 vs. stream 2 to carry heat per kelvin:

$$CP_1\left(\vartheta_{1,\alpha} - \vartheta_{1,\omega}\right) = CP_2\left(\vartheta_{2\omega} - \vartheta_{2,\alpha}\right)$$

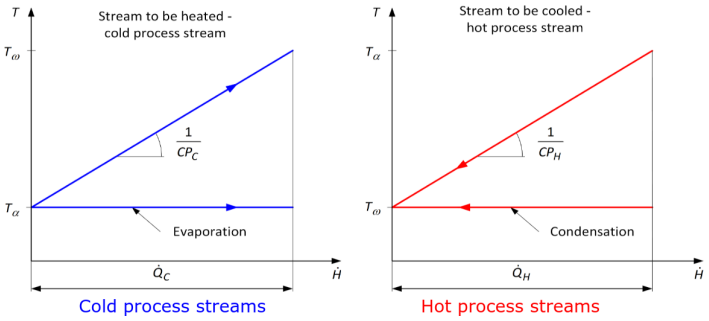
$$R_1 = \frac{CP_1}{CP_2} = \frac{\vartheta_{2,\omega} - \vartheta_{2,\alpha}}{\vartheta_{1,\alpha} - \vartheta_{1,\omega}} = \frac{\Delta\vartheta_2}{\Delta\vartheta_1}$$

T,H diagram

Cold and Hot streams

The process is extracted into material streams that must be heated (heating need) or cooled (cooling need):

Cold streams	Hot streams
Heating needed (heat sink): • Warming of milk for pasteurisation • Heating a heat transfer oil • Preparation of hot water • Evaporation of fruit juice	Cooling needed (heat source) • Cooling of a salt solution • Cooling a polymer melt • Cooling water from air compressors • Chiller refrigerant • Flue gas

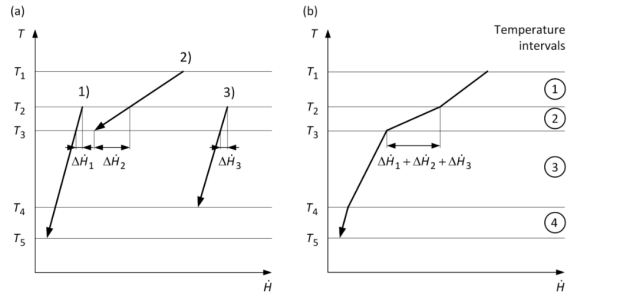


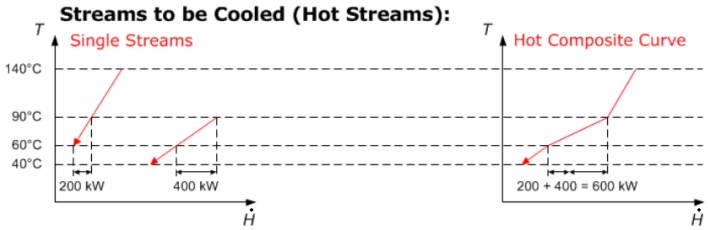
Slope of the streams

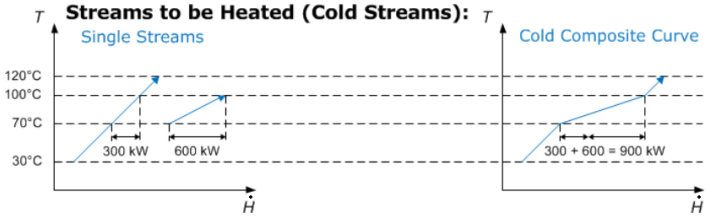
$$\frac{\Delta T}{\Delta \dot{H}} \Longleftrightarrow \frac{\Delta T_1}{CP_1 \cdot \Delta T_1} = \frac{1}{CP_1} \Longleftrightarrow \frac{\Delta T_2}{CP_2 \cdot \Delta T_2} = \frac{1}{CP_2}$$

Composite curve

$$\frac{1}{CP_n} = \frac{\Delta T_n}{\Delta \dot{H}_n} = \frac{\Delta T_n}{\sum_i CP_i \Delta T_n} = \frac{1}{\sum_i CP_i}$$







II. Energy and Cost Targets

ΔT_{min}	HR Potential	Required Utilities	HEX Area	Investments Costs	Energy Costs
decreases	increases	decreases	increases	increases	decreases
increases	decreases	increases	decreases	decreases	increases

Cost targets

Calculation of Annual Total Costs (Annualized Total Costs)

$$C_{\text{tot}} = AC_{\text{inv}} + C_{\text{Op}}$$

Annuity factor

$$A = \frac{Z(1+Z)^n}{(1+Z)^n - 1}$$

Z = interest rate
n = Depreciation/Amortisation period

Investment costs

$$C_{\text{Inv}} = C_{\text{HEN}} + C_{\text{ECU}} + C_{\text{Ind}}$$

HEN = Heat exchanger network cost
ECUs = Energy conversion units
Ind = Independent (eg. control integration)

Capital costs for heat exchanger network (HEN)

$$C_{\text{HEN}} = \sum C_{\text{HEX}} = \sum \left(a + C_b \left(\frac{Q}{Q_b} \right)^m \right)$$

C_{HEX}	Capital cost of heat exchanger (HEX)
a	Base cost
C_b	Cost of reference heat exchanger
Q_b	Heat transfer area of reference heat exchanger
m	Degression exponent
Q	Area of HEX to be costed

Q refers to quantity (in this case, Area for the HEX costs)

Annual operating costs

$$C_{\text{Op}} = C_{\text{En}} + C_{\text{M}} + C_{\text{Per}}$$

Maintenance costs

$$C_{\text{M}} = C_{\text{Inv}} f_{\text{M}}$$

Personnel costs

$$C_{\text{Per}} = C_{\text{Inv}} f_{\text{Per}}$$

Annual energy costs

Annual energy costs are the sum of the annual hot and cold utility costs and the energy costs of the ECUs.

$$C_{\text{En}} = C_{\text{HU}} + C_{\text{CU}} + C_{\text{En,ECU}}$$

Hot utility costs

$$C_{\text{HU}} = \dot{Q}_{\text{HU}} \tau c_{\text{HU}}$$

Cold utility costs

$$C_{\text{CU}} = \dot{Q}_{\text{CU}} \tau c_{\text{CU}}$$

$\dot{Q}_{\text{HU/CU}}$	Hot/cold utility requirements (heat flow rate)
$C_{\text{HU/CU}}$	Annual energy cost of hot/cold utility
τ	Annual operating hours
$c_{\text{HU/CU}}$	Specific energy cost of hot/cold utility

Energy costs of energy conversion units (ECUs)

$$C_{\text{En,ECU}} = \sum E_{\text{ECU}} c_{\text{En}}$$

$C_{\text{En,ECU}}$	Annual energy cost of ECUs
E_{ECU}	Annual energy requirement of an ECU
c_{En}	Specific cost of the energy supplied to an ECU

Heat exchanger network cost target

Assuming an equal distribution of the total heat transfer area among the required heat exchangers:

$$C_{\text{HEN,Target}} = N_{\text{Units,Target}} \left(a + C_b \left(\frac{A_{\text{HEN,Target}}}{N_{\text{Units,Target}} Q_b} \right)^m \right)$$

Single-stream heat transfer area target

$$A_{\text{HEN,Target}} = \frac{\dot{Q}}{\Delta T_m} \left(\frac{1}{\alpha_h} + \frac{1}{\alpha_c} \right)$$

Heat transfer area target (Uniform Bath Formula)

$$A_{\text{HEN,Target}} = \sum_{k=1}^{K} \frac{1}{\Delta T_{m,k}} \left[\sum_{i=1}^{I} \frac{\dot{Q}_{i,k}}{\alpha_i} + \sum_{j=1}^{J} \frac{\dot{Q}_{j,k}}{\alpha_j} \right]$$

k, K	Enthalpy interval index k through K
i, j	Hot and cold stream indices within interval k
I, J	Number of hot and cold streams within interval k
α_i, α_j	Heat transfer coefficients of hot and cold streams
$\dot{Q}_{i,k}, \dot{Q}_{j,k}$	Heat flow rates of the streams within interval k
$\Delta T_{m,k}$	Mean temperature difference within interval k
$A_{\text{HEN,Target}}$	Target total heat transfer area of the HEN

Number of heat exchangers

$$N_{\text{Units,Target}} = [S_{\text{above}} - 1] + [S_{\text{below}} - 1]$$

$N_{\text{Units,Target}}$	Target number of heat exchangers
S_{above}	Number of streams including utilities above the pinch
S_{below}	Number of streams including utilities below the pinch

Energy targets

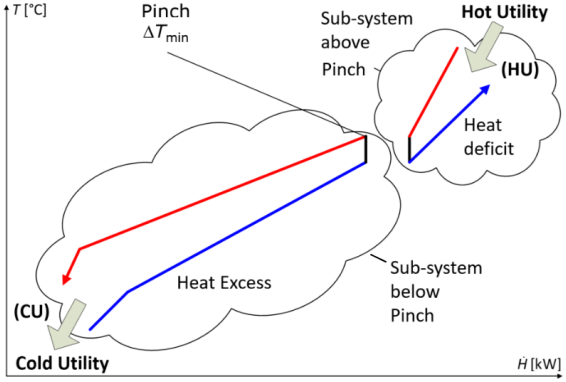
Three main pinch rules

1. No heat transfer through the pinch
2. No external heating below the pinch
3. No external cooling above the pinch

Characteristics of the pinch point

The pinch divides the total system into two subsystems with:

- Heat deficit above the pinch
- Heat excess below the pinch.

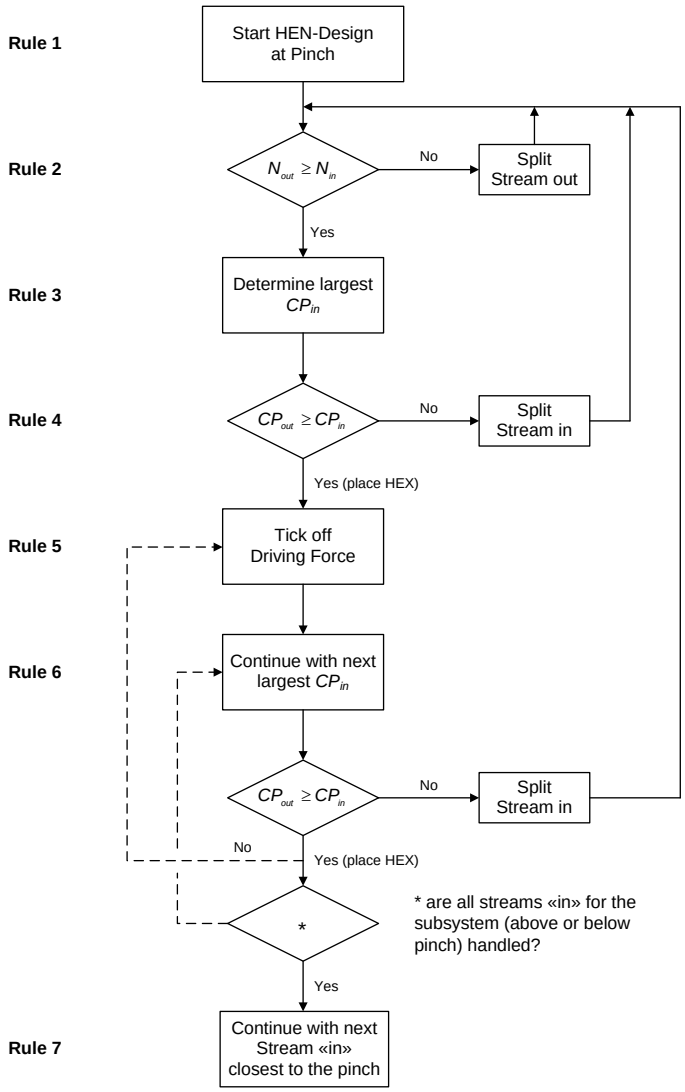


III. Heat Exchanger Network (HEN)

MER = Maximum Energy Recovery (for HR)
MER = Minimum Energy Requirement (for CU/HU)

Inwards = Stream that is approaching the PT
Outwards = Stream temperature that is leaving the PT

Rules for the design of HENs



These design rules help to achieve the targets.

1. Start from the pinch and progress outwards
2. Number rule: $N_{\text{out}} \geq N_{\text{in}}$: every stream which heads in the direction of the pinch has a match partner in a HEX
3. Begin with the largest CP_{in} : stream which heads in the direction of the pinch with the largest CP_{in}
4. CP rule: $CP_{\text{out}} \geq CP_{\text{in}}$: no HEX less than ΔT_{min}
5. HEX Optimization:
   a) **"Tick Off Rule"**: minimize the number of HEX and/or
   b) **"Driving Force Plot"**: minimize the area of the HEN
6. Continue with the largest CP_{in} at the Pinch
7. If all HEXs at the Pinch are set, then proceed with the stream closest to the Pinch that leads to the Pinch

IV. Process requirements

TODO

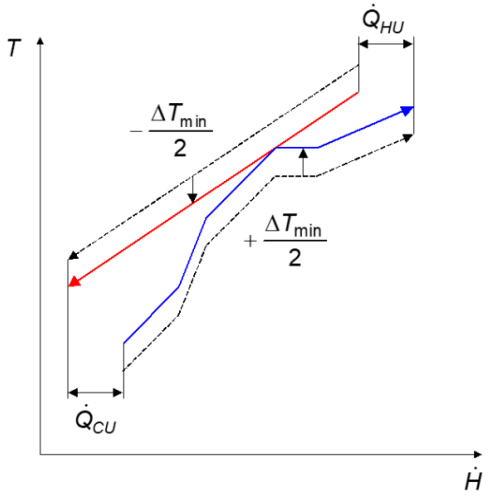
V. Optimizing energy supply systems (ESS)

Optimizing utilities ESS

Grand Composite Curve (GCC)

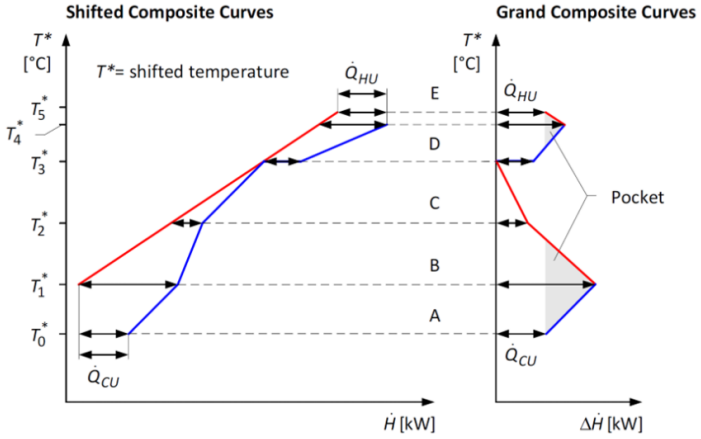
GCC is formed from the Composite Curves (CC) and applies only for a certain ΔT_{min}.

To construct the GCC, the CCs are each shifted vertically by $\pm\frac{\Delta T_{\text{min}}}{2}$ until they touch at the pinch point.



The horizontal distance correspond to the required hot and cold utilities.

The horizontal distance between the shifted CCs are transferred into another $T, \dot{H}$-diagram



Pockets in the GCC

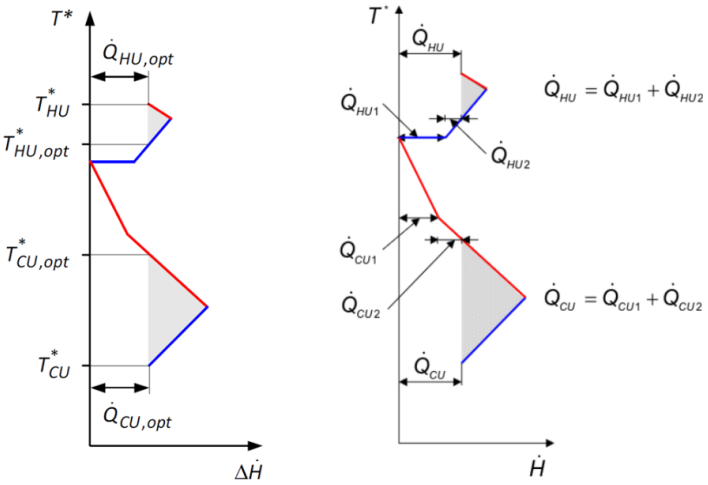
Pockets show internal heat recovery. Heat released at higher temperatures supplies process heat demand at lower temperatures, without external utilities.

Optimization of the utility system temperature levels

Calculation of the real temperature of the hot utility from the optimized shifted temperature, according to the GCC:

$$T_{HU} = T^*_{HU} + \frac{\Delta T_{\text{min}}}{2} \quad ; \quad T_{CU} = T^*_{CU} - \frac{\Delta T_{\text{min}}}{2}$$

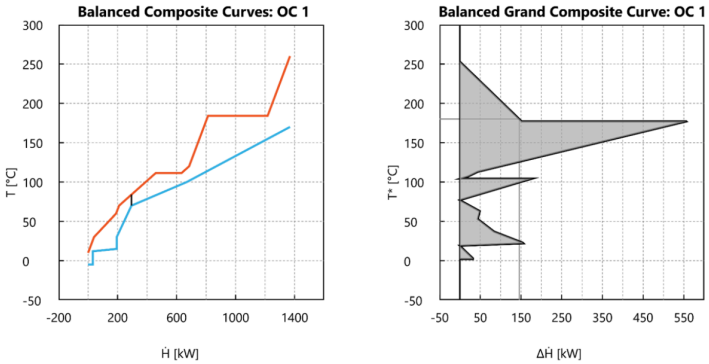
Optimal temperature levels



Utility optimization and consequences for the HEN design

HEN has to be constructed on the CCs including the utilities
→ Balanced Composite Curve (BCC)

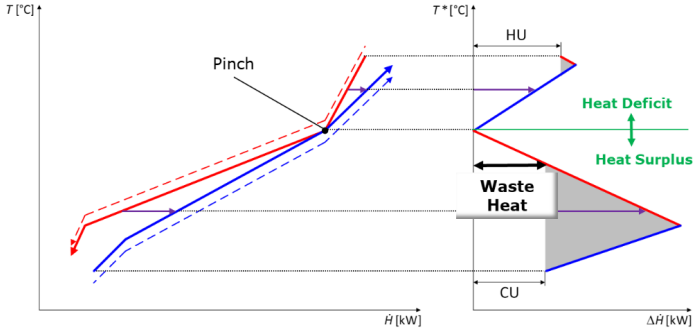
Balanced Composite Curves (BCCs) and BGCCs



1. The pinch rules must be applied at the Utility-Pinch
2. Between two pinches is a clash of the pinch rules
3. Always start at the most constaint pinch

VI. Integrating ECUs

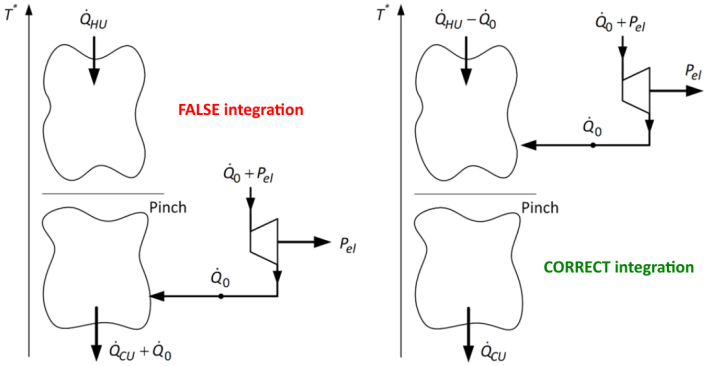
ECU = Energy Conversion Unit



Combined Heat and Power (CHP) systems

- Thermal engines generate electrical energy and heat
- η_{el} 30-40%, η_{tot} 80-90%
- Typical CHP: gas turbine, steam turbine, combustion engine

Integrating CHP



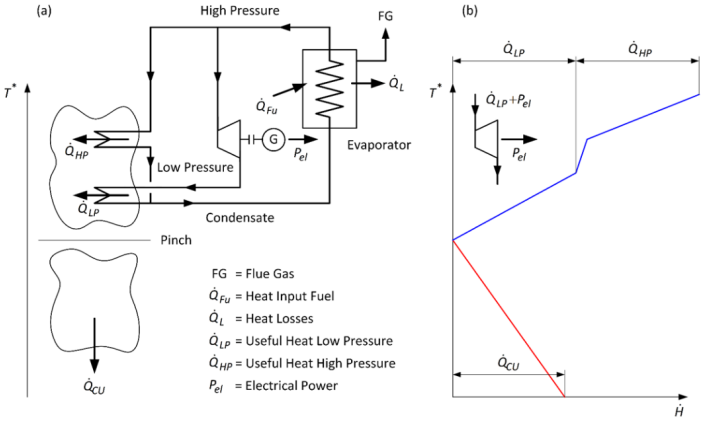
False integration – Below the Pinch

The heat surplus is increased by the amount of heat generated by the CHP. This results in a higher cooling requirement.

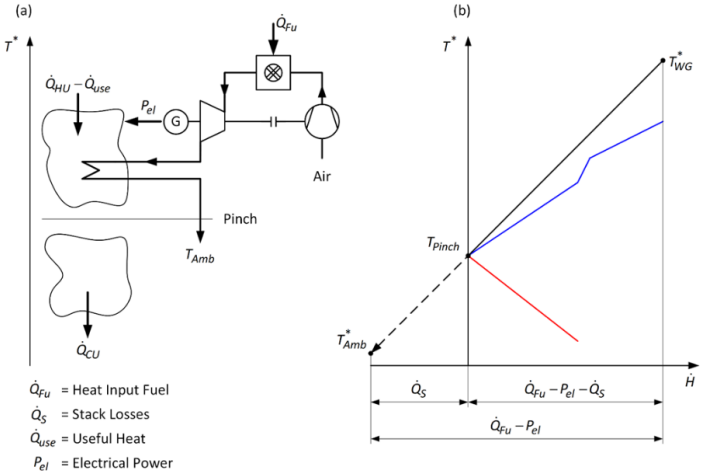
Correct integration – Above the Pinch

The heat deficit is decreased by the amount of heat generated by the CHP. This results in a decrease of the heating requirement.

Method to integrate a steam turbine



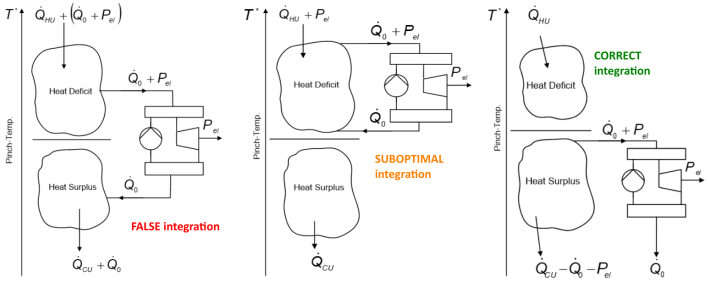
Method to integrate a gas turbine



Organic Rankine Cycle (ORC) Systems

Basics:

1 → 2: adiabatic (polytropic) compression
2 → 3: isobaric heat input (high temperature level)
3 → 4: adiabatic expansion
4 → 1: isobaric heat output (low temperature level)



False integration – Over the Pinch

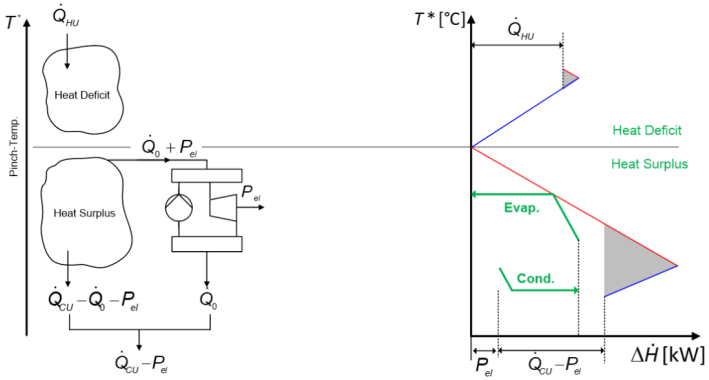
The heat deficit is increased by the amount of electrical power generated and condenser heat. The heat surplus is increased by the amount of condenser heat. Both a higher heating and cooling requirement results.

Suboptimal integration – Above the Pinch

The heat deficit is increased by the amount of electrical power generated leading to a higher heating requirement

Correct integration – Below the Pinch

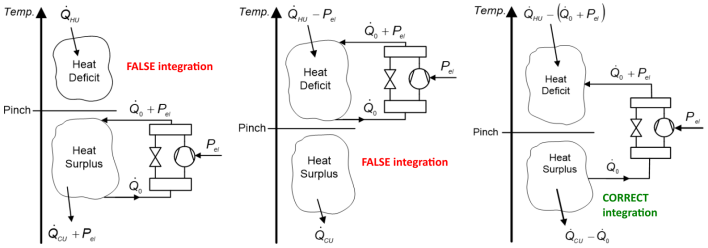
The ORC reduces the cooling demand of the process and the associated operating costs.



Integrating Heat Pumps (HP)

Basics:

1 → 2: adiabatic (polytropic) compression
2 → 3: isobaric heat output (high temperature level)
3 → 4: adiabatic expansion
4 → 1: isobaric heat input (low temperature level)



False integration – Below the Pinch

The heat surplus is increased by the amount of compressor electrical power leading to a higher cooling requirement.

False integration – Above the Pinch

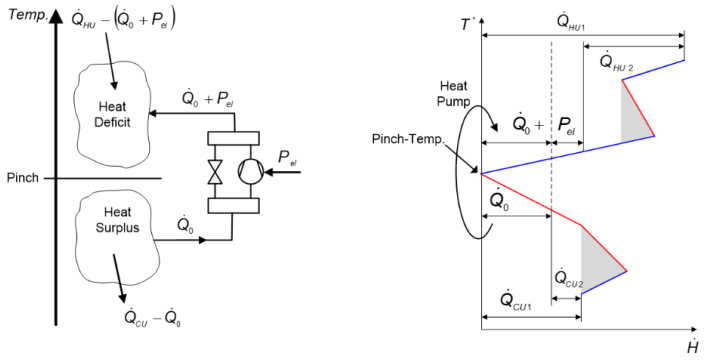
The heat deficit is reduced by the amount of compressor electrical power. From an energetic perspective this type of integration is equivalent to electrical heating.

Correct integration – Over the Pinch

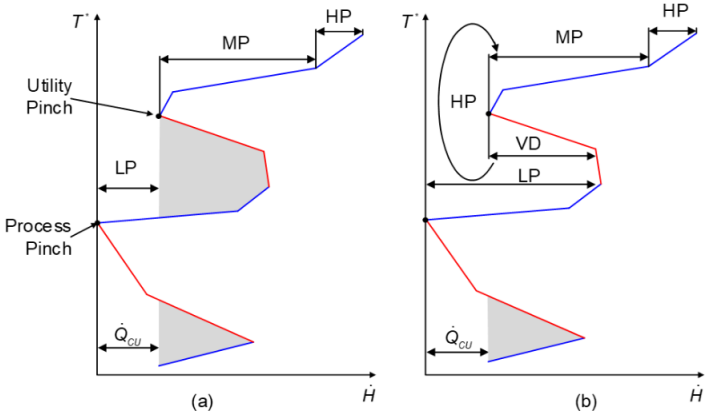
The heat pump reduces both the cooling and the heating demands of the process and the associated operating costs.

Correct integration and basic design using the GCC

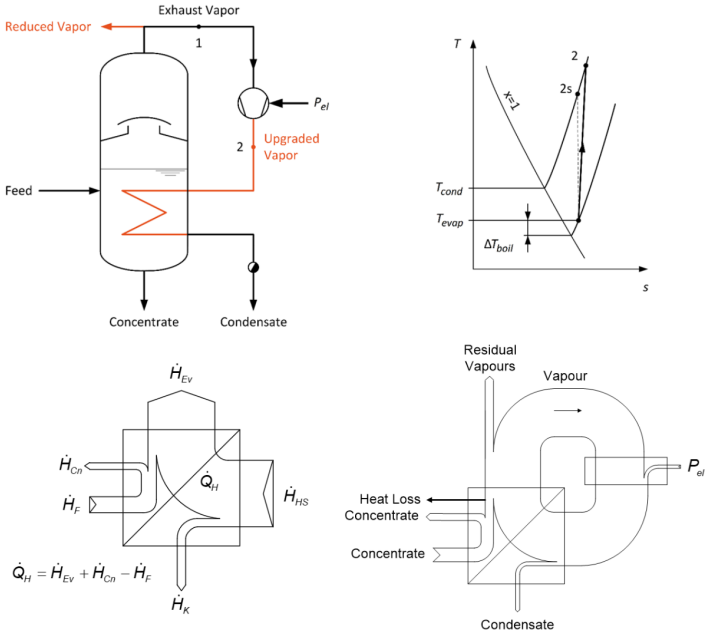
A heat pump may only absorb heat below the pinch (excess heat) and give off heat above the pinch (heat deficit) to the process.



HP integration across the Process Pinch or Utility Pinch



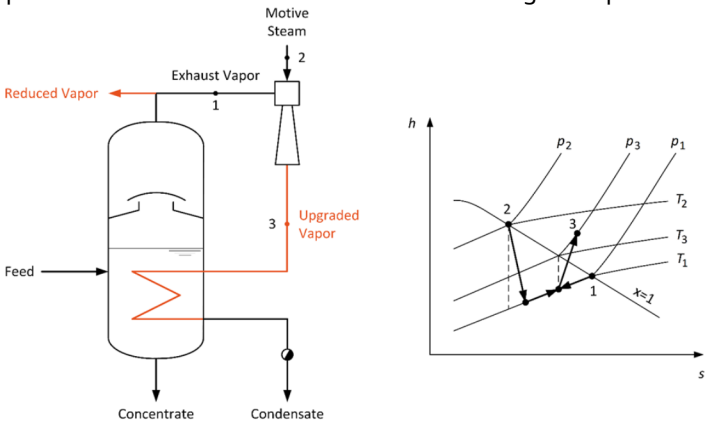
Principle of Mechanical Vapor Recompression (MVR)

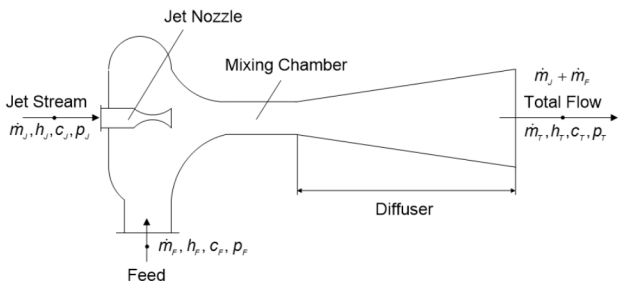


The drive energy of the compressor is usually less than 10% of the heat when supplied by direct heating.

Principle of Thermal Vapor Recompression (TVR)

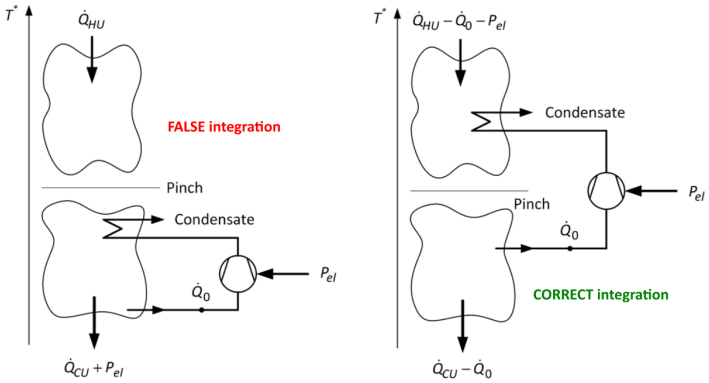
A typical application of an TVR is for the compression of exhaust steam to a higher pressure





The resulting enthalpy of the resulting mixture of the two steam flows is typically between 3 to 6 times higher the enthalpy of the exhaust steam! This fact makes a TVR suitable for different energetic situations than an MVR.

Integration of MVR/TVR

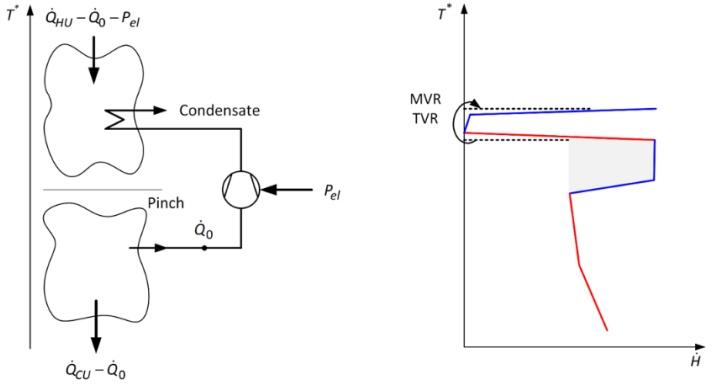


False integration – Below the Pinch

The heat surplus is increased by the amount of compressor electrical power leading to a higher cooling requirement.

Correct integration – Over the Pinch

The MVR/TVR reduces both the cooling and the heating demands of the process and the associated operating costs.



A vapor recompression may only absorb heat below the pinch (excess heat) and give off heat above the pinch (heat deficit).